
🌐 AWS Route 53

✅ What is Amazon Route 53?

Amazon Route 53 is a **scalable and highly available Domain Name System (DNS) web service** designed to give developers and businesses a reliable way to:

- **Route end users to internet applications**
- **Manage domain names**
- **Monitor the health of resources**

> 🔹 It works as **DNS + Domain Registrar + Health Check service**.

🔎 Why is it Called "Route 53"?

- 53 = TCP/UDP port number used by DNS.

🧱 Core Components

Component	Description
Hosted Zone	Container for DNS records for a domain
Record Set	DNS entries (e.g., A, CNAME, MX)
Health Checks	Monitor endpoint health

Traffic Flow	Advanced routing policies
Domain Registration	Buy and manage domains
Resolver	Forward and conditional DNS for hybrid cloud (Route 53 Resolver)

🌍 Types of Hosted Zones

Type	Purpose
Public Hosted Zone	Resolves domain on the **internet**
Private Hosted Zone	Resolves domain names within a **VPC** (internal DNS)

🧭 Routing Policies (Record Routing Types)

1. 🎯 **Simple Routing**
    - One record = One resource.
    - No health check support.
    - Used when you want to resolve to a **single static resource** (like an S3 website).
2. 🔁 **Weighted Routing**
    - Split traffic based on **weights** (e.g., 70% to one server, 30% to another).

    - Useful for **A/B testing** or **gradual rollout**.
3. 🌐 **Latency-Based Routing**
    - Direct users to the **region with lowest latency**.
    - Uses health checks and Route 53's global latency measurements.
4. 🧠 **Failover Routing**
    - Primary–secondary setup.
    - Route 53 checks **health** and fails over if primary is down.
    - Good for **disaster recovery**.
5. 🧩 **Geolocation Routing**
    - Route traffic based on user's **country/continent/region**.
    - Useful for **regional laws**, **licensing**, **language preferences**.
6. 🗺️ **Geoproximity Routing (with Traffic Flow)**
    - Route based on **location + bias** (increase/decrease traffic flow to a region).
    - Requires traffic flow policies.
7. 🧾 **Multi-Value Answer Routing**
    - Route to **multiple healthy resources** (max 8).

- Can work like simple load balancing.

📄 Common DNS Record Types

Record	Description
A	IPv4 address
AAAA	IPv6 address
CNAME	Alias for another domain
MX	Mail servers
TXT	Arbitrary text (e.g., domain ownership, SPF)
NS	Name servers for domain
Alias	AWS-specific version of CNAME for **root domain** (e.g., alias for s3-website)

> 🟢 **Alias records** are **free** (CNAME-like), support **naked domain** (e.g., mydomain.com), and only work inside AWS.

🔒 DNS + Security (Route 53 + IAM + CloudTrail)

- You can restrict who manages hosted zones and records via **IAM policies**.
- **CloudTrail** logs DNS changes for auditing.
- Supports **DNSSEC (Domain Name System Security Extensions)** for DNS validation.

🔍 Route 53 Health Checks

- Can monitor:
    - **HTTP**, **HTTPS**, **TCP**
    - **CloudWatch Alarms**
- Can be associated with **failover routing**.
- Automatically reroutes if health check fails.

🏢 Route 53 Resolver (Hybrid DNS)

- Used to **connect on-premises networks to AWS DNS**.
- Two components:
    - **Inbound endpoint** – for on-premises to query AWS DNS.
    - **Outbound endpoint** – for AWS to query on-premises DNS.

🔄 Useful in hybrid cloud, VPC peering, or AWS Direct Connect scenarios.

💰 Pricing Summary

Feature	Pricing
Hosted Zone	$0.50/month per hosted zone

Queries	$0.40–$0.60 per million queries (depending on region)
Health Checks	$0.50/month per health check
Domain Registration	Depends on TLD
Traffic Flow	$50/policy/month

🛠️ Use Cases

1. Hosting website using **S3 + Route 53**
2. Setup **Failover DNS** between AWS and on-prem
3. Route traffic based on **geographic region**
4. Use **private hosted zones** for internal microservices
5. Integrate **Route 53 Resolver** with Direct Connect/VPN
6. Implement **multi-region high availability**

🖼️ Example: Failover Routing Diagram

```
┌──────────────┐
 │  Route 53  │
 └──────┬───────┘
       │
┌───────▼───────┐
│ Primary ALB│ <── Health Check ──┐
```

```
└──────┬───────┘          │
   │           Failover
┌──────▼───────┐          │
│  Secondary   │ <── Health Check ◄────┘
└──────────────┘
```

🧠 **Tips**

- Use **Alias Records** instead of CNAME for AWS services.
- Always use **health checks** with failover and multi-value routing.
- Enable **logging** and **DNSSEC** for security compliance.
- Use **Private Hosted Zones** for microservices inside VPCs.
- Use **Route 53 Resolver** for hybrid environments.
